

Deepak Sathyan

Curriculum Vitae

[arXiv](#) | [InspireHEP](#) | [Google Scholar](#) | [ORCID](#) | [Website](#)

✉ dsathyan@tamu.edu

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland

College Park, MD

Ph.D. in Physics

Aug 2018 – July 2024

Thesis: Unifying Searches for New Physics with Precision Measurements of the W

Boson Mass

Advisor: Dr. Kaustubh Agashe

Boston University

Boston, MA

Bachelor of Arts in Physics, Summa Cum Laude with Honors

Aug 2014 – May 2018

GPA: 3.9

Minor: Mathematics

Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Experience

Postdoctoral Research Associate

College Station, TX

Mitchell Institute for Fundamental Physics and Astronomy

Sept 2024 – Aug 2027

Texas A&M University

3 years

Research Interests

Beyond the Standard Model searches with Collider Physics

searches at the LHC, sub-GeV dark matter, invisible new physics via precision measurements, Kaluza-Klein modes in Randall-Sundrum models, baryon number violation models, muon collider phenomenology

Searches for sub-GeV Dark Matter

Constraints on dark matter-neutrino interactions, dark photon production at the LHC, dark sector production at muon collider, astrophysical signals of cosmic ray-dark matter scattering, astrophysical sources of sub-GeV dark matter

Neutrino Physics

Supernova neutrino attenuation and detection, neutrino oscillations in open systems, neutrinophilic mediators

Publications

5. New laboratory constraints on neutrinophilic mediators

July 2024

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

- 4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W**

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

Apr 2024
- 3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET**

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)

Oct 2023
- 2. Energy-peak based method to measure top quark mass via B-hadron decay lengths**

K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)

Dec 2022
- 1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension**

K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Aug 2020

Preprints

- 4. Intensity-Frontier Signals of Warped Extra Dimensions**

D. Kim, D. Sathyan, A. Verma
arxiv.org/abs/2608.20482

Aug 2026
- 3. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering**

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010

May 2026
- 2. A Baryon and Lepton Number Violation Model Testable at the LHC**

A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064

Aug 2025
- 1. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis**

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000

July 2025

Articles in Preparation

- 1. LHC as a Beam Dump: Probing Light Dark Photons**

B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
arxiv.org/a/sathyan_d_1

Sept 2026
- 2. New Probes of Dark Sectors at a Muon Collider**

B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
arxiv.org/a/sathyan_d_1

Jan 2027

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| <p>3. Exploring open systems approach to neutrino oscillations
 A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
 arxiv.org/a/sathyan_d_1</p> | <p>Mar 2027</p> |
| <p>4. LHC Signals for KK Gravitons Producing 8-Particle Final States
 K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
 arxiv.org/a/sathyan_d_1</p> | <p>Mar 2027</p> |

Snowmass2021 Contributions ---

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| <p>5. TF07 Snowmass Report: Theory of Collider Phenomena
 F. Maltoni et al.
 arxiv.org/abs/2210.02591</p> | <p>Oct 2022</p> |
| <p>4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021
 K. Agashe et al.
 arxiv.org/abs/2209.11267</p> | <p>Sept 2022</p> |
| <p>3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021
 T. Bose et al.
 arxiv.org/abs/2209.13128</p> | <p>Sept 2022</p> |
| <p>2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond
 K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
 arxiv.org/abs/2204.02928</p> | <p>Apr 2022</p> |
| <p>1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models
 K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
 arxiv.org/abs/2203.13305</p> | <p>Mar 2022</p> |

Muon g-2 Articles ---

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| <p>3. The straw tracking detector for the Fermilab Muon g-2 Experiment
 B.T. King et al.
 arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)</p> | <p>Nov 2021</p> |
| <p>2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab
 Muon g-2 Collaboration
 arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)</p> | <p>Apr 2021</p> |
| <p>1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm
 Muon g-2 Collaboration
 arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)</p> | <p>Apr 2021</p> |

Ph.D. Thesis

- 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Seminars

- 7. CMS Exotica general meeting** May 2024
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
- 6. Seminar at Texas A&M** Dec 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 5. Seminar at UT Austin** Dec 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 4. CMS SUSY general meeting** Nov 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 3. HEP Seminar at University of Pittsburgh** Nov 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 2. HEP Seminar at Northwestern University** Oct 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 1. Theory Seminar at Washington University in St. Louis** Oct 2023
A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

- 14. Light Dark World 2026 at Carleton University** July 2026
Can the LHC be sensitive to the Light Dark World?
- 13. CETUP* Workshop** June 2026
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering
- 12. Particle Physics on the Plains at the University of Kansas** Nov 2025
A Baryon and Lepton Number Violation Model Testable at the LHC
- 11. CETUP* Workshop** June 2025
Can the LHC be sensitive to light dark mediators?

10. Phenomenology Conference at the University of Pittsburgh Can the LHC be sensitive to light dark mediators?	May 2025
9. Particle Physics on the Plains at the University of Kansas A comprehensive analysis of supernova neutrino-dark matter interactions	Nov 2024
8. Texas TACOS @ UT Austin A comprehensive analysis of supernova neutrino-dark matter interactions	Oct 2024
7. NuFact 2024 Conference at Argonne National Laboratory A comprehensive analysis of supernova neutrino-dark matter interactions	Sept 2024
6. Mitchell Conference at Texas A&M University A comprehensive analysis of supernova neutrino-dark matter interactions	May 2024
5. Particle Physics on the Plains at the University of Kansas A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	Oct 2023
4. Phenomenology Conference at the University of Pittsburgh Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos	May 2023
3. Phenomenology Conference at the University of Pittsburgh Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)	May 2022
2. April APS Meeting Signals of KK graviton from extended warped extra dimensions at the LHC (II)	Apr 2021
1. Phenomenology Conference at the University of Pittsburgh Signals of KK graviton from extended warped extra dimensions at the LHC (II)	May 2020

Conferences Organized

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Texas A&M University, May 13-16, 2025

Awards

Breakthrough Prize in Fundamental Physics
Muon g-2 Collaboration

Apr 2026

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions

Jan 2023 – Jan 2023

TASI 2024: The Frontiers of Particle Theory

June 2024 – June 2024

Outreach

BU Student Mentorship Program PRISM (2016-2018)

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

TAMU Physics and Engineering Festival (2025-2027)

Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" (December 2024)

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules